Unfunded Mandates and Judicial Performance: Evidence from Alcohol Prohibition

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Abstract

While criminal legal reforms are often enacted to protect vulnerable populations, developing countries struggle to put changes into practice as reforms get implemented as 'unfunded mandates' that overwhelm existing judicial infrastructure. This paper evaluates the effect of one such mandate—alcohol prohibition in a large Indian state—on judicial performance. I analyse the effects of the ban using a difference-in-differences research design using a neighboring state as a control due to shared geography and historical institutions, combined with micro-level data on 1.1M court cases. I find that the reform increased demands on the justice system, worsened case resolution, but did not affect the quantity of daily work performed by judges. The findings highlight how top-down criminal legal reforms can inadvertently degrade the quality of justice.

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1 Introduction

Legal reforms have the potential to spur development—which explains the large push toward understanding what reforms work. While introducing good, appropriate policy measures to improve societal outcomes is important, the implementation of programs may fail if additional responsibility is devolved to public institutions without increasing capacity. Such "unfunded mandates" increase the workloads for state officials and have far reaching consequences on the performance of public institutions.

Particularly, substantive criminal legal reforms determine the limits of acceptable "legal" and "illegal" behaviors within a society—by changing the definitions of crimes and the severity of punishments. By broadening the criminal code to include previously unregulated activities, criminal legal reforms expand the state's mandate in enforcing certain behaviors. Without commensurate investments in institutional capacity, such reforms can fundamentally undermine the performance of the criminal justice system.

This paper examines the institutional consequences of one such unfunded mandate by evaluating the effect of total prohibition of alcohol in a large Indian state—Bihar. While lawmakers enacted the policy to address societal harms, the ban criminalized any production and consumption of alcohol within the state. People continued to violate the law, overwhelming the courts with thousands of new cases.

I assess how courts were affected by the increased workload, and argue that the reform impeded the functioning of the judiciary. The flood of cases increased judicial workloads, and since the reform was an unfunded mandate, it expanded the scope of the criminal code without providing the institutional support necessary to manage the resulting surge in litigation. While judges maintained a similar amount of daily work after the reform, it significantly worsened case resolution and increased pendency across the affected courts.

Empirically identifying the effects of the reform is challenging since the timing of the ban could be endogeneous, due to the presence of nationwide trends that could affect judicial performance—so a simple before-and-after comparison would fail to isolate the impact. To address this concern, I utilize a difference-in-differences design leveraging courts in Jharkhand, a neighboring state, which did not pass the reform. Jharkhand shares common historical institutions and geographic characteristics, which makes for a reasonable control group to overcome the absence of a clean counterfactual. I assemble a dataset of over 1.1 million cases filed in the District & Sessions Courts in the two states, that allows me to measure court performance at a granular level.

First, I document that the reforms increased demands on the justice system. I find that the ban on alcohol indeed increased workloads in the judiciary, particularly driven by an increase in case filings. Next, in line with the theoretical expectations, I find that the reform led to large consequences on judicial performance—while the reform did not change the number of hearings conducted by judges, it decreased case resolution by 13%.

Further, I eliminate alternate explanations that relates to shifts in judicial capacity. To show this, I construct a novel dataset of biographical information of judges presiding in and recruited to both states during the study period. My analysis confirms that the reform did not lead to significant changes in composition or the overall number of judges serving in the two courts. That is, given that the judicial capacity in the two states remained constant, the decline in performance in Bihar can be attributed to the surge of cases on account of the alcohol ban. Possible mechanisms (to be tested) may point to re-prioritization of the pending cases, demonstrated by an increase in hearings on bail cases (which are taken up on priority) and due to a decrease in time to work on and write judgements.

These findings demonstrate that criminal legal reforms can induce severe strain on scant judicial resources, with unintended downstream consequences. These findings are in line with previous studies that find that juggling tasks worsens judge productivity (Coviello, Ichino, and Persico 2015). While this study focuses on a reform that expands the criminal code, the theoretical framework extends to other public sector contexts wherein mandates exceed capacities, typical of states in the developing world (Dasgupta and Kapur 2020).

This paper makes important contributions by examining how unfunded mandates can undermine the justice system. In doing so, it advances the literature on unfunded mandates, that has mainly looked at fiscal pressures on lower levels of governments (Posner 1998; Rodríguez-Pose and Vidal-Bover 2024; Ross 2018a). A small body of scholarship has explored these mandates in the context of developing countries (Shi and Ye 2024; Khambule 2020). While unfunded mandates across levels of government has been noted as reason for budget constraints in India (Rodden, Eskeland, and Litvack 2003), this paper expands the conceptual scope to include horizontal institutional impacts. Specifically, it analyzes how legislative actions can reconfigure the functions of allied state institutions—–such as the judiciary.

Second, I contribute to the scholarship on the determinants of judicial and bureaucratic performance. Research in this area frequently attributes under-performance to capacity constraints (Rao 2024; Muralidharan, Niehaus, and Sukhtankar 2016) and incentives (A. Q. Khan, Khwaja, and Olken 2019; A. Q. Khan, Khwaja, and Olken 2016). This paper shifts focus to the demand-side pressures created by the legal environment. By analysing a major expansion of the criminal code, I demonstrate how the nature and complexity of the law can drive judicial performance.

Finally, this paper contributes to the growing body of empirical literature evaluating alcohol prohibition in India. While existing scholarship has examined the ban's impact and noted improved public health indicators (Chakrabarti et al. 2024; Krishnatri and Vellakkal 2026), reduction violence against women (Chakrabarti et al. 2024; Dixit and Mukherjee 2025), reduction in crime (Suryanarayana et al. 2023; Mukherjee and Kumari 2026), increase in undertrial prisoners (P. Kumar and Anupama 2023), and reduced (but not eliminated) alcohol use (Balhara et al. 2023; Sharma 2023). The institutional consequences for the legal system have remained largely unexplored, although legal scholars have pointed to the burdens on the criminal justice system (Choudhary and Prakash 2025). I contribute to this literature by assessing the downstream effects of the policy on the judiciary.

2 Theory

Unfunded mandates are responsibilities devolved to a public institution which are not accompanied by necessary resources to fulfill it (Rodríguez-Pose and Vidal-Bover 2024). Such mandates have been typically studied in federalist systems—wherein higher levels of government mandate costly functions on lower levels without providing financial support.

Such mandates have been found to affect the fiscal structure of states and was a common phenomenon in the United States, which put in place the "Unfunded Mandates Reform Act" in 1995 to avoid the federal shifting of costs to state or local government. In practice, states have called out the costs associated with unfunded federal mandates as recently as 2016 (Ross 2018b).[1]

Unfunded mandates are not unique to the United States—and indeed, may be a common outcome of the decentralization process (Rodríguez-Pose and Vidal-Bover 2024). This phenomenon is observed in other countries as well, although by different names. It is referred to as "cost-shifting" in Australia and "service responsibility downloading" in Canada (Basdeo 2012). Such mandates affect the fiscal responsibility of lower levels of government, who raise additional revenue from alternate sources to balance the shortfall (Shi and Ye 2024).

While the idea of unfunded mandates has been previously applied to mandates given *across* levels of government (typically in federalist systems), I argue that unfunded mandates can also be directed *between* horizontal government agencies—wherein one branch increases the work of another branch, without concurrent increases in funding. This paper studies a phenomenon of horizontal unfunded mandates—wherein enactment of new laws in the legislature increases the work of allied state institutions without a commensurate increase in capacity, leading to operational constraints. In this context, legislatures pass popular but unenforceable laws that are merely "symbolic" (Tallent, Jan, and Sattelmayer 2026). The horizontal dimension of unfunded mandates captures a hidden cost of governance, particularly in developing states.

Criminalizing formerly legal behaviors places additional burdens on law enforcement and judicial systems through a mandatory change. By designating a new crime, the state expands the breadth of activities that need to be regulated, redressed and punished by the state. Since the state has created a new threshold for arrest and prosecution, the unfunded mandate surges cases against offenders in the court system. The defendants may seek to avoid pre-trial detention, that creates a secondary "demand-side shock" on the courts in the form of bail petitions. This idea motivates my first hypothesis:

[1] In fact, at least 18 states have amended their constitutions or enacted laws to limit the states ability to issue such unfunded mandates (Ross 2018b).

H_1: Expansion of the criminal code leads to a surge in case filings

While these increases might be driven by criminal cases, the resulting surge creates a deterrent for private litigants. As the court congestion and delay becomes visible to the public, would-be litigants face increased costs (in the form of expected delays) to enter and pursue formal litigation. This discourages the filing of new civil disputes between private parties, as potential plaintiffs anticipate that their cases will be de-prioritized in an overwhelmed system.

H_{1a}: Expansion of the criminal code leads to a decrease in civil case filings

The state's institutions, facing increased workloads, can react to such constraints by either selectively enforcing new laws or reducing the quality of service. Such constraints become acute in the judicial system. Courtrooms are stressful places, and the evolving institutional and societal environment require judges to constantly learn and adapt. The complexities of the modern legal codes increases the demand on judges, who routinely face traumatic cases that affect the performance of their occupational duties (Chamberlain and Miller 2009) and face occupational stress (Flores et al. 2007). These stresses arise from a number of factors. First, judges primarily (but not exclusively) engage in highly specialized and cognitively demanding activities that have additional administrative responsibilities on top of the caseload (Gunawardena, O'Donnell, and Williamson 2019). Large backlogs may lead to difficult and unrealistic targets set by their senior counterparts, further adding to the stress experienced by judges. The shortage of staff and human resources only aggravates this issue. Judges also encounter political interference, especially in cases involving high stakes. Politicians can use the bait of transfers, promotions, and post-retirement positions to make judges comply with their expectations (Aney, Dam, and Ko 2021).

Unfunded mandates directly expand the volume of litigation on a judge's active docket, forcing a reallocation of limited time and attention. Organizational literature establishes that such workload shocks negatively affect job performance and accuracy (Siswanto et al. 2019; Janib et al. 2021; Bruggen 2015; Sherf, Venkataramani, and Gajendran 2019; Cox-Fuenzalida 2007), particularly when workers must perform multiple tasks simultaneously (Cox-Fuenzalida and Angie 2005). For the judiciary, this "task juggling" is especially detrimental: as judges juggle multiple tasks due to changes in their workload, it dimishes their ability to dispense with their portfolio of cases (Coviello, Ichino, and Persico 2015).

This phenomenon mirrors the "overload" experienced by bureaucrats in resource-constrained environments. Facing operational capacity constraints, officials shift focus away from long-term planning and execution towards immediate, particularistic tasks (Dasgupta and Kapur 2020). Given that bureaucrats have limited physical and mental resources, the state of "overload" affects bureacratic performance—since they are unable to execute projects effectively due to rationing their time in sub-optimal ways. Increasing workloads can affect the performance of judges in similar ways—it reduces the time available to write judgments (Gunawardena, O'Donnell, and Williamson 2019) and affects their decision-making ability (Rachlinski and Wistrich 2017).

To evaluate these theoretical expectations, I distinguish between two core components of judicial performance that focus on work performed by judges (Staats, Bowler, and Hiskey 2005): (1) effectiveness: which determines the degree to which decisions are made and enforced in a timely manner and (2) efficiency: which is related to maximizing outputs given a set of inputs.

While related dimensions, there can be tradeoffs between the two. maximizing performance in one dimension may come at the cost of performance in another dimension. While efficiency views judicial activity as a production process that involves producing maximum output given an input, efficacy concerns the ability of a judicial system to match the demands of justice (i.e. solving cases promptly). An efficient judiciary may not necessarily be effective: slow judiciaries may be able to use their resources more efficiently, or fast judiciaries may be inefficiently using their resources (Marciano, Melcarne, and Ramello 2019).

I hypothesize that the reform worsens the ability of judges to make decisions in a timely manner. As the backlog of cases increases without any change in existing work hours or judges, it is not possible to execute the additional amount of work with the same resources. Judges may prioritize cases that can be resolved quickly or need to be heard urgently. This comes at the cost of hearing the more complex cases that require more time and coordination of various agencies to move to a resolution. Thus, cases that remain on the judge's docket face greater pendency. This leads to my second hypothesis:

H_2: Expansion of the criminal code worsens clearance capacity of courts

Despite the decline in case resolutions, the effort exerted by judges likely remains stable. A judge's daily schedule is dominated by the time that they spend in conducting hearings. Judges actively participate in scheduling hearings for new and existing cases. While deciding which cases are to be heard, judges consider formal rules on case timelines—for instance, cases regarding bail need to be taken up immediately as they concern individual liberties. Judges also consider their performance incentives—the metrics based on which their performance is evaluated by their superiors—while deciding which and how many cases to hear. Given the strong metrics-based incentives for judges, I argue that there will be no change in the amount of routine work performed by judges, as measured by case hearings.

H_3: Expansion of the criminal code does not lead changes in the amount of work performed by judges (as measured by hearings)

Although the theory highlights how horizontal unfunded mandates can worsen the performance of state institutions, it does not apply to institutions that have discretion or authority to refuse the additional work given their operational constraints. Further, while presence of hierarchical oversight assures that the daily expected work is performed, institutions with little or no oversight could reduce overall work performed as officials become overwhelmed. Finally, while the theory concerns the demand and supply side changes in terms of volume of work performed, it does not account for changes in the quality of work that is executed.

3 Context

Long considered a societal vice, alcoholism is a medical condition wherein a person cannot control their alcohol use and consumption—with adverse social, occupational, or health consequences. While alcohol sale forms a large base of excise revenue, several Indian states have instituted varying levels of alcohol prohibition to combat the issues associated with alcoholism—particularly those pertaining to its effects on domestic violence. And not without reason—a relationship exists between alcohol consumption and women reporting domestic violence (S. Kumar and Prakash 2016; Cafferky et al. 2018). Alcohol prohibition in Indian states typically targets the manufacturing, sale and consumption of liquor within state borders (Dar and Sahay 2018).

The latest state to pass such a reform was Bihar, one of India's poorest state. Since 2005, Bihar has been led by Chief Minister Nitish Kumar at the helm. During the Bihar State Assembly Elections in 2015, the incumbent Chief Minister announced a plan to ban alcohol consumption in the state if he were re-elected, citing the prevalence of alcoholism and domestic violence. After winning the election, he kept his election promise by passing legislation to this effect, despite bearing heavy losses in excise revenues. The prohibition law was passed in April 2016, which banned the manufacture, sale and consumption of liquor within the state borders. Further, stringent jail terms and fines were stipulated for offenders.

Research on alcohol prohibition policies in developing countries shows that bans on alcohol are associated with lower levels of domestic violence (Luca, Owens, and Sharma 2015), and indeed in Bihar, the ban led to a lower reporting of sexual assault and harassment against women (Khurana and Mahajan 2022). A. Roy (2020) finds that alcohol bans across Indian states reduced certain kinds of crime, and the ban in Bihar particularly reduced the incidence of violent crime (Suryanarayana et al. 2023). While the decrease in crime was driven by increase in police capacity (Suryanarayana et al. 2023), no such increase in judicial capacity was made during the period to handle the increasing work load. Thus, the alcohol ban is an appropriate setting to study the effects of an unfunded mandate that increased workloads without increasing capacity of the judiciary.

The Indian judiciary is headed by the Supreme Court is at the apex level, which is often touted as the 'most powerful judiciary in the world'. Further, there are 24 High Courts (encompassing states or larger regions), and each district has a District and Sessions Court, staffed by senior judges. Further, subordinate courts hear both civil and criminal cases. Judges are recruited based on their performance in competitive judicial examinations at junior levels and directly from the bar for some senior positions—the norms for which vary across states. Judge in the District and Sessions Courts can hear both civil and criminal cases.

The justice system in India is plagued with low capacity—there are 20 authorized judge positions per million—however, this number is much lower once vacancies are accounted for (Rao 2022). This is a fifth and a tenth of the positions as compared to the US and UK, respectively.[2] Given that the capacity of courts is not frequently revised, courts are not equipped to cope with large demand-side shocks.

4 Data and Empirical Strategy

The case-level data used in this paper comes from the eCourts project in India, which undertook large scale digitization of Indian courts capturing the universe of all digitized cases. As a result of this initiative, cases have been digitized and uploaded online with rich metadata, that includes details about the nature of the case, litigators, lawyers and judges who heard the case, dates of filing and decision, along with outcomes.

I start with assembling detailed case-level data by scraping individual case records for all cases filed in the lower courts in Bihar and Jharkhand to augment the data publicly shared by Ash et al. (2025). Using this data, I create a court-room year dataset for the District & Sessions Courts in both states.[3]

Using the case-level data, I construct three outcomes of interest. I calculate the total number of filings in each court-year for the period between 2015 and 2018, which helps me test H_1, and further, the total number of filings by case type (either civil or criminal) to test $H_{1(a)}$. Next, for each case I calculate whether it was decided within one year of the filing date. I take averages of this for all cases in each court-year period, which gives me the probability of decision within the same year for each court-year to test H_2. Third, I for each case, I use the hearings data to count the number of hearings conducted in each court-year, which

[2]Rao explains: "The total number of judge posts and judge recruitment drives in a district are determined jointly by the respective state high court and the state-level executive (through budget allocation) There is no clear rule on how the number of judge posts are revised over time, and periodic reports by the Law Commission of India, an executive body under the central government Ministry of Law and Justice (particularly, the Law Commission Report No.245) point out that this is relatively ad hoc without any specific calculus. Typically, the numbers are determined at the time of setting up a court's physical infrastructure, which happens once every few decades (mainly at the time of district formation)." (Rao 2022)

[3]As per Bihar's Excise (Amendment) Act of 2016 (the alcohol ban reform), the servere punishments stipulated meant that cases would only be tried at the highest level courts. This mandate was written into the law which states that the Sessions Judge has the power to pass any sentence authorized by the Act.

Table 1: Comparing Bihar and Jharkhand

	Bihar	Jharkhand
Land Area	79,714 sq. km.	94,163 sq. km.
Capital	Patna	Ranchi
Districts	37	18
Population (in millions)	82	26
Sex-Ratio (2001)	921	941
Deccadal Growth Rate (91-01) (%)	28.43	23.19
Population Density (2001)	880	338
Percentage Literacy (%)	47	54
Percentage Female Literacy (%)	33	39
% Hindu	82	68
% Sarnaism (Indigenous Religion)	1	13
% Scheduled Castes	16	12
% Scheduled Tribes	1	26

Notes: This table compares characteristics of Bihar and Jharkhand, which are largely similar. Jharkhand has a larger population of Scheduled Tribes and those who practice indigineous religions.

allows me to test H_3. In addition to the court-room dataset, I also collect a novel dataset of biographical information of judges presiding in and recruited in Bihar and Jharkhand, to rule out alternate explanations in line with the identification strategy that I describe below.

My main goal is to estimate the causal effect of the alcohol ban on the judiciary. The main hypothesis is that the alcohol ban *worsened* outcomes related to judicial performance. The target estimand is the difference between judicial performance of a court in which the alcohol ban is implemented and the judicial performance of a court had the policy *not* been implemented. A natural way to conduct such an analysis is to leverage a difference-in-differences design, where I use the data from a control group to impute untreated outcomes in the treated group. In other words, using the control group helps us learn about the unobserved counterfactual.

The analysis compares courts in Bihar with courts in its neighboring state—Jharkhand. Jharkhand was carved out of the territories of Bihar in 2000. The two states were a part of a combined orginal state before 2000 and share the same history, institutions, and geography. Indeed, Jharkhand has been leveraged as a suitable control group for Bihar in similar strategies used in several other papers (Dixit, Mukherjee, and Rajan 2023; T. Khan 2022; Muralidharan and Prakash 2017; Suryanarayana et al. 2023).

Bihar was reorganized in 2000 based on the demands from indigenous communities (Scheduled Tribes) over fifteen years (A. K. Roy 2000). As seen in Table 1, the demographics of Bihar and Jharkhand differ in a few ways—particularly, the population density and the cultural make up. Jharkhand has a much larger tribal population and a larger sex-ratio.

The key identifying assumption is that the change in court outcomes from pre- to post reform in Jharkhand is a good proxy for the counterfactual change in untreated potential outcomes in Bihar. I argue this is the case since both the states were under the same institution (the Patna High Court) which was only divided in 2003. Thus, it is more appropriate to compare the judicial characteristics of Bihar and Jharkhand, in addition to the overall comparisons made in Table 1. Judges used to be recruited from a common pool—and indeed many of these judges (who were recruited before the split) were still a part of the judicial service during the period analysed. Further, all results test for the presence of trends in the pre-treatment period.

The estimand is the average treatment effect for the treated courts (ATT). I use `FEct`—particularly, the *Two-way Fixed Effects Counterfacutal* to make counterfactual predictions (Liu et al. 2022) for courts that do not experience the alcohol ban. The counterfactual estimator is one in which Y_{it} (that is, the outcome Y for court i at time t without treatment) is imputed based on a two way fixed effects model:

$$Y_{it} = \delta \mathbf{D}_{it} + \alpha_i + \eta_t + \epsilon_i \tag{1}$$

Here, δ is the parameter of interest, $\mathbf{D}_{it}$ indicates the treatment which equals 1 if court i is observed post reform at time t, α_i indicates court fixed effects, while η_t indicates time fixed effects. Standard errors are clustered at the court x time level. The `FEct` estimator is preferred to the standard difference-in-differences estimator as it is better equipped to visualize and test pre-trends (Liu et al. 2022). The use of `FEct` is also driven by the rationale that we expect heterogeneous treatment effects over time—the effects of the alcohol ban on judicial performance in the first year might be more muted than its effect in later years.

5 Main Results

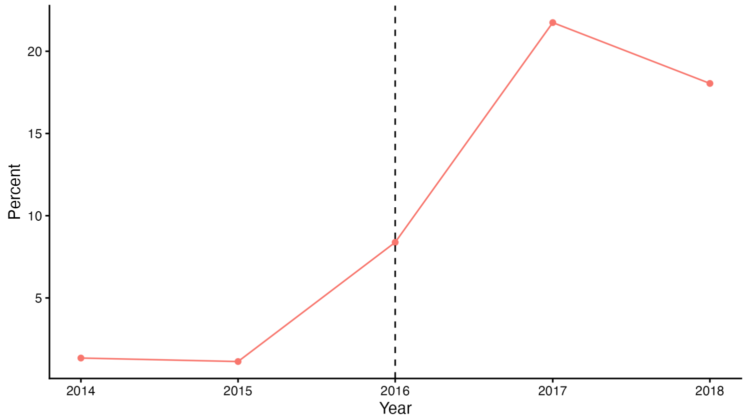
I first use data from the National Family and Health Survey to show that while the alcohol ban decreased alcohol consumption, it did not completely eliminate it. I focus the analysis only for men, since women's consumption of alcohol in Bihar is less than 2%. From the results in appendix (to be added), we see that while the alcohol consumption in Bihar for men was about 30% before the ban, the ban reduces alcohol consumption among men in Bihar by 14%. This indicates that there still remain a considerable number of men who are potentially offenders and can be arrested. Further, the decrease of 14% is a ceiling effect, since the ban might be affecting people's reporting of alcohol-consumption as well. Respondents would be less likely to report illegal activity, hinting at a larger pool of offenders than estimated in the survey. All this points to a significant number of offenders who continue to consume alcohol post the state-wide ban, who can be arrested for violating the prohibition rule.

Next, I test H_1 to see whether the alcohol ban led to an increase in the number of case filings in the District and Sessions Courts. Since a new category of crime has been created, an overall increase in filings may be observed. On the other hand, as litigants observe large demand increases on courts that they are unable to meet, the tendency to get involved in litigation may decrease.

Using the case-level data, I first plot the number of cases related to the alcohol ban in Figure 1. We see an increase in filings under the new act in Bihar in and after 2016. At the peak in 2017, over 20% of all filings in District and Sessions Courts were related to the alcohol prohibition.

I test this descriptive result more systematically using the difference-in-differences analysis, with the number of yearly filings as an outcome. The main finding from this analysis is that the number of overall filings increases in each year from 2016 to 2018. On average, the number of filings in the Disrict and Sessions courts in Bihar increase by an average of 1386 cases in the post-treatment period, as seen in Figure 2. This corresponds to a 65% increase

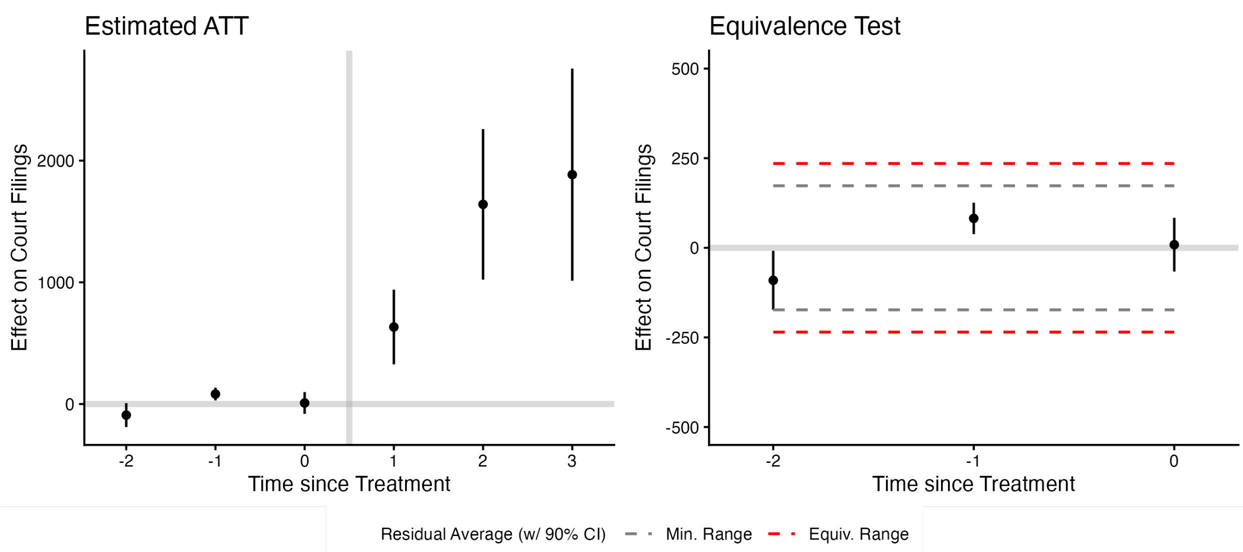
Figure 1: Filings pertaining to prohibition in Bihar courts



Notes: Plot shows the share of filings related to the alcohol ban as a percent of all cases filed in the District and Sessions Courts of Bihar.

in case filings as compared to the pre-treatment period.

Figure 2: Effect of prohibition on case filings in courts



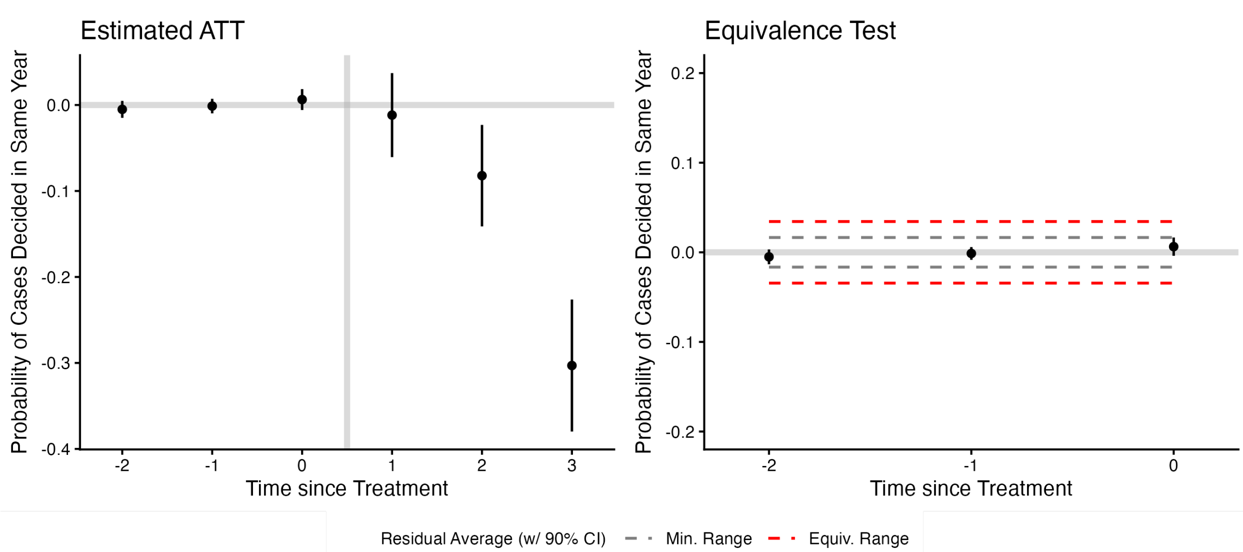
Notes: Left plot shows the dynamic treatment effect estimates for case filings before and after the alcohol ban. Bars denote the 95% confidence intervals from 200 bootstraps clustered by court. Right plot shows the pre-treatment average prediction errors and 90% confidence interval. Red dashed lines denote the equivalence bounds. Black dashed lines mark the minimum range bound.

While a visual inspection of the trend does not show any change in outcome during the pre-treatment period, I use the equivalence test to check for the presence of a pre-trend. Here, I check whether the 90% confidence intervals for the estimated ATTs in the period prior to the treatment fall within a pre-specified minimum range. This would indicate that courts in the two states are not diverging in outcomes prior to the reform. The p value of the equivalence test is less than 0.000, which indicates a high confidence that equivalence holds. In the figure, the black dotted lines indicate the minimum range within which the null of inequivalence can be rejected, which is within the equivalence range.

This is evidence that judges in Bihar indeed faced a demand-side shock in the form of increased filings—wherein they had to cope with increased workloads. I will also show that the increase in filings is driven particularly by criminal cases in the appendix (to be added), and that I do not find evidence of a similar increase in the filing of civil litigation.

Next, I test H_2: whether the ban decreased worsend case resolution and pendency in the justice system. I exclude bail cases for this analysis since they are typically prioritized based on a principle of justice. Figure 3 plots the estimated effect of the alcohol ban on probability that cases within the court are decided in the same year. We see a gradual decrease starting 2016 (the year of the reform) on the probability of cases being decided in the same year as filed. We see the largest decrease (in magnitude) in 2018—overall, the mean probability of cases reaching a decision drops by 13%.

Figure 3: Effect of prohibition on cases decided in the same year

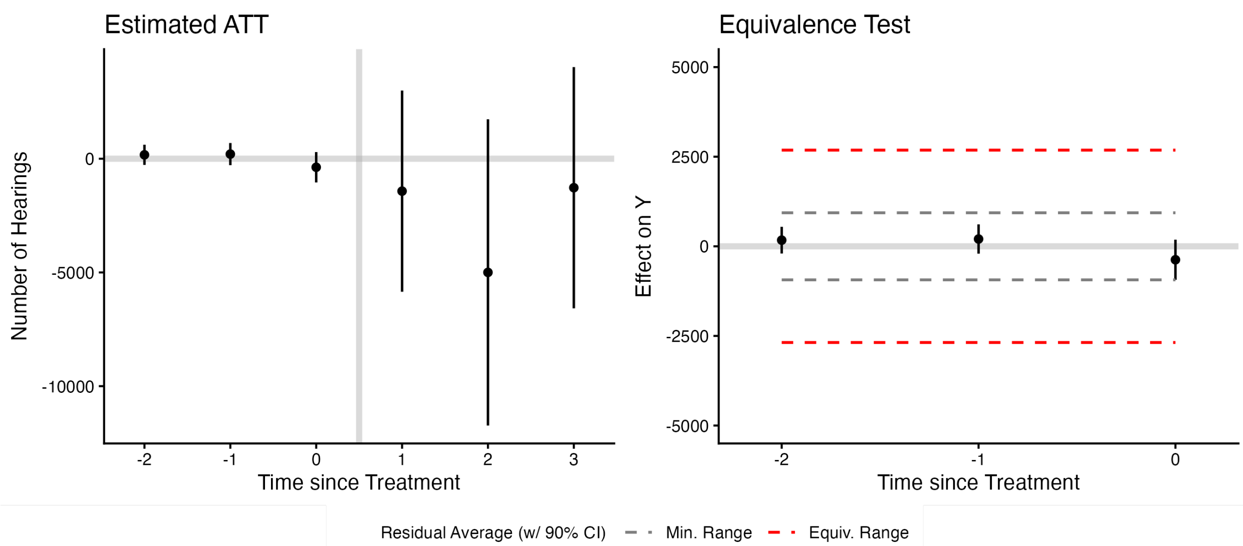


Notes: Left plot shows the dynamic treatment effect estimates for probability of case decision before and after the alcohol ban. Bars denote the 95% confidence intervals from 200 bootstraps clustered by court. Right plot shows the pre-treatment average prediction errors and 90% confidence interval. Red dashed lines denote the equivalence bounds. Black dashed lines mark the minimum range bound.

I perform the same equivalence test as before to check for the presence of a pre-trend. I check if the 90% confidence intervals for the estimated ATTs in the pre-treatment period exceed the equivalence range. Since the p value is smaller than 0.000, I conclude that that equivalence holds. On average, the alcohol ban decreases the probability of getting a decision in the same year in Bihar by 13%. This is statistically and substantively a large effect and can have large downstream implications on the efficacy of the justice system in Bihar during this period.

My final empirical goal is to understand the effect of the alcohol ban on the daily work performed by judges by turning to the number of hearings conducted. Figure 4 shows that while I do not find presence of pre-trends, I do not see evidence that the alcohol ban affects the number of hearings conducted by judges.

Figure 4: Effect of prohibition on hearings conducted by judges

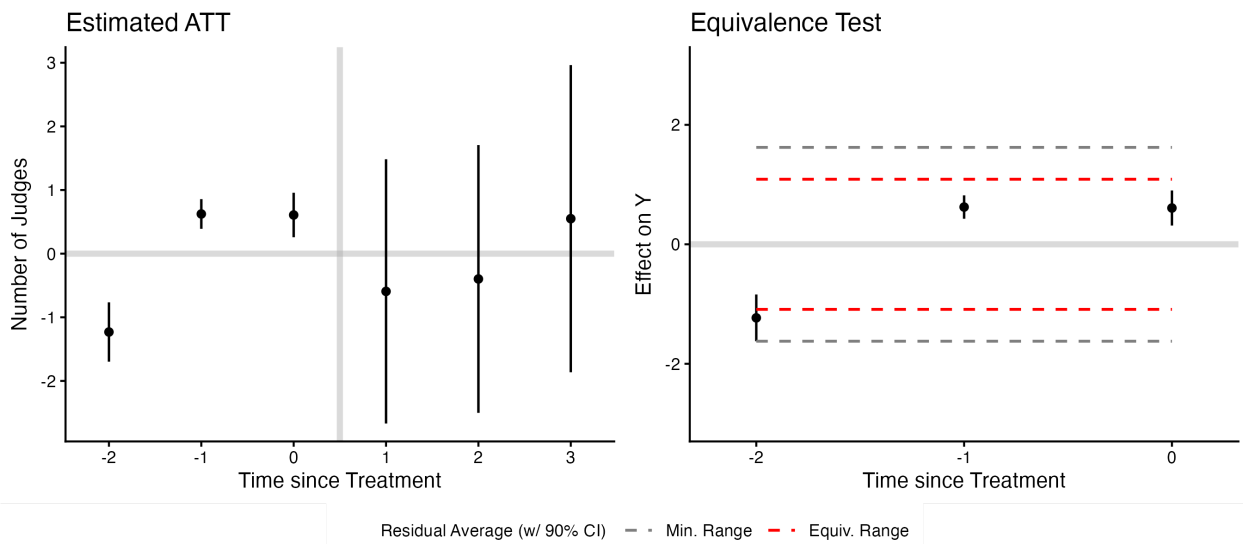


Notes: Left plot shows the dynamic treatment effect estimates for the number of hearings conducted before and after the alcohol ban. Bars denote the 95% confidence intervals from 200 bootstraps clustered by court. Right plot shows the pre-treatment average prediction errors and 90% confidence interval. Red dashed lines denote the equivalence bounds. Black dashed lines mark the minimum range bound.

5.1 Reform does not lead to changes in capacity

Can the decrease in judicial performance be explained by changes in judicial capacity? In order to rule out this alternate explanation, I construct a dataset on the number of all judges serving in the Bihar and Jharkhand judiciary during each time period. Figure 5 shows that while the set-up passes the equivalence test with a p value of 0.004, there are significant pre-treatment trends before the reform is enacted. While we see there is no evidence of an increase or decrease in judicial capacity Bihar after the reform, I will sub-set the data to only focus on the number of judges serving in the District & Sessions Courts to fully rule this out as an alternate explanation.

Figure 5: Effect of prohibition on the number of judges serving in all courts



Notes: Left plot shows the dynamic treatment effect estimates for the number of judges serving in all courts before and after the alcohol ban. Bars denote the 95% confidence intervals from 200 bootstraps clustered by court. Right plot shows the pre-treatment average prediction errors and 90% confidence interval. Red dashed lines denote the equivalence bounds. Black dashed lines mark the minimum range bound.

6 Mechanisms

In this section, I discuss the main mechanisms I plan to test. Judges serving in courts where the reform is in place could be changing the mix of cases they are hearing. As their daily workloads increase, they may be prioritizing the hearings and decisions for bail cases (which are seen as more urgent) at the cost of other cases. This motivates an analysis of heterogeneous treatment effects based on case types. In order to conduct this, I compute separate outcomes for civil, criminal and bail cases for each court, and test whether the alcohol ban has a differential effect based on case types. This change in case mix is the main mechanism that can explain the change in case pendency and delays.

Another reason that explains why cases experience delays is because judges find it harder to find time to write out the final decision for the case. To analyse this, I look at whether the alcohol ban affects the time between the final hearing and the date on which the decision is made. This might be suggestive evidence to show that it is the reform makes it harder for judges to schedule time to write decisions, possibly hinting at a decrease in judicial quality.

Finally, an important source of heterogeneity is the enforcement of the reform. There are two places where the alcohol ban would be most strongly enforced. First, in districts with a larger share of MLAs from the ruling party, and second, in districts that are more urban, and closer to the state capital. In order to evaluate this, I plan to create two new variables. First, I compute the share of ruling parties of MLAs in all districts, and present an analysis restricting the treatment districts to districts which had greater than 50% MLAs belonging to the ruling party. Second, I restrict the analysis only to urban and border districts to check whether the impact of the ban is different for rural, urban and border districts, which can point to heterogeneity based on the degree of enforcement.

7 Conclusion

In this paper, I evaluate the effects of an unfunded legal reform on the judiciary—I study the effect that a state-wide alcohol ban on the judicial system. While literature focusing on unfunded mandates has largely focused on the US context on fiscal constraints applied across government levels, this paper examines and extends this idea in two ways. First, it argues that unfunded mandates can be directed between horizontal government agencies, and second, that these mandates can lead to organizational constraints. In the case examined in the paper, a law-making body increases the workload of the judicial system through criminalization of behaviors, without adequately increasing capacity to handle and perform the additional work.

Leveraging a difference-in-differences design, I find that the ban on alcohol in Bihar leads to massive demand-side shock through an increase in filings, which worsens case resolution without changing the number of hearings conducted by judges. I also rule out alternate explanations pertaining to judicial capacity and argue that the main mechanism that leads worsens outcomes is the change in the case-mix which leads to re-prioritization—judges prioritize cases that involve a bail request due to strong norms regarding their disposal, at the cost of other pending cases.

Unfunded mandates require further study. While this paper looks at a particular form of the mandate which seeks to criminalize behaviors, future work could differentiate from effects of unfunded mandates that seek to decriminalize behaviors. More broadly, different mechanisms can drive the effects of policy changes on institutions that do not require the state to make costly investments for implementation.

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